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MAMBO mapping of Spitzer c2d small clouds and cores[⋆,⋆⋆,⋆⋆⋆]

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ABSTRACT

Aims. To study the structure of nearby (<500 pc) dense starless and star-forming cores with the particular goal to identify and understand evolutionary trends in core properties, and to explore the nature of Very Low Luminosity Objects ($\leq 0.1\ L_{\odot}$; VeLLOs).
Methods. Using the MAMBO bolometer array, we create maps unusually sensitive to faint (few mJy per beam) extended ($\approx 5'$) thermal dust continuum emission at 1.2 mm wavelength. Complementary information on embedded stars is obtained from Spitzer, IRAS, and 2MASS.
Results. Our maps are very rich in structure, and we characterize extended emission features ("subcores") and compact intensity peaks in our data separately to pay attention to this complexity. We derive, e.g., sizes, masses, and aspect ratios for the subcores, as well as column densities and related properties for the peaks. Combination with archival infrared data then enables the derivation of bolometric luminosities and temperatures, as well as envelope masses, for the young embedded stars.
Conclusions. Starless and star-forming cores occupy the same parameter space in many core properties; a picture of dense core evolution in which any dense core begins to actively form stars once it exceeds some fixed limit in, e.g., mass, density, or both, is inconsistent with our data. A concept of *necessary conditions* for star formation appears to provide a better description: dense cores fulfilling certain conditions *can* form stars, but they do not need to, respectively have not done so yet. Comparison of various evolutionary indicators for young stellar objects in our sample (e.g., bolometric temperatures) reveals inconsistencies between some of them, possibly suggesting a revision of some of these indicators. Finally, we challenge the notion that VeLLOs form in cores not expected to actively form stars, and we present a first systematic study revealing evidence for structural differences between starless and candidate VeLLO cores.

Key words. stars: formation – ISM: evolution – ISM: structure – ISM: dust, extinction – ISM: clouds

1. Introduction

Stars form from dense gas ($\gg 10^5$ cm^{-3}). Such gas is, e.g., found in discrete nearby ($\lesssim 500$ pc) cold (≈ 10 K) small-scale ($\lesssim 0.1$ pc) condensations referred to as dense cores (Myers et al. 1983). These are thought to be the sites where low-mass stars form (Myers & Benson 1983). Their properties provide the initial conditions for star formation. It is thus necessary to understand the physical state of dense cores in order to be able to understand the star formation process in detail.

Some of the many physical parameters of dense cores are their sizes, masses, and densities. Given that thermal dust continuum emission at ≈ 1 mm wavelength is a better mass tracer than molecular emission lines (e.g., Tafalla et al. 2002), and that modern bolometer cameras enable complete maps of the dust emission from dense cores to be made within a short time, dust emission maps are a prime tool to study the above core parameters. Dust emission surveys thus enable a systematic overview of dense core properties.

In recent years several surveys studied dense cores in large (up to several 10 pc) and massive (up to several $10^3\ M_{\odot}$) complexes of molecular clouds (Motte et al. 1998; Johnstone et al. 2000, 2001, 2004, 2006; Johnstone & Bally 2006; Hatchell et al. 2005; Enoch et al. 2006, 2007; Young K. E. et al. 2006; Stanke et al. 2006). Most cores in these clouds are in groups and clusters (e.g., Enoch et al. 2007). Although it is believed that most stars in the galaxy form in such environments (Magnani et al. 1995), two issues make it difficult to study the fundamental physics of star formation in such environments. First, cores can be confused in crowded regions forming clusters; it becomes difficult to identify and study individual cores (see, e.g., Motte et al. 1998; see also Ward-Thompson et al. 2007, for illustrative examples). Second, cores might interact in crowded regions; more parameters than relevant in isolated cores influence the evolution of dense cores. These problems can be avoided by studying dense cores in relative isolation (Clemens & Barvainis 1988; Bourke et al. 1995) instead of cores hosted in larger (up to $\gtrsim 100$ pc) complexes of giant molecular clouds (Blitz 1993).

The present study therefore focuses on isolated cores. We have observed both starless cores and cores with embedded stars,

⋆ Based on observations with the IRAM 30-m telescope.
⋆⋆ Figure 1, Tables 3–5 and Appendices A, B are only available in electronic form at http://www.aanda.org
⋆⋆⋆ Data used for the maps (FITS files) are only available in electronic form at the CDS via anonymous ftp to
cdsarc.u-strasbg.fr (130.79.128.5), or via
http://cdsweb.u-strasbg.fr/cgi-bin/qcat?J/A+A/487/993

Appendix A: Dust emission properties

This appendix summarises formulas to convert the observed flux density of dust emission into column densities and masses. These are evaluated for the standard assumptions for dust properties made by the c2d collaboration. The discussion presented here is intended to serve as a future reference for the c2d project.

A.1. Molecular weight

As one usually wishes to express column densities in terms of particles per area, the molecular weight needs to be introduced into the equations. The molecular weight per hydrogen molecule, μ_{H_2}, is defined via $\mu_{H_2} m_H \mathcal{N}(H_2) = \mathcal{M}$; here $\mathcal{M}$ is the total mass contained in a volume with $\mathcal{N}(H_2)$ hydrogen molecules, and m_H is the H-atom mass. It can be calculated from cosmic abundance ratios. For hydrogen, helium, and metals the mass ratios are $\mathcal{M}(H)/\mathcal{M} \approx 0.71$, $\mathcal{M}(He)/\mathcal{M} \approx 0.27$, and $\mathcal{M}(Z)/\mathcal{M} \approx 0.02$, respectively, where $\mathcal{M} = \mathcal{M}(H) + \mathcal{M}(He) + \mathcal{M}(Z)$ (Cox 2000). As $\mathcal{N}(H) = 2\mathcal{N}(H_2)$, $\mathcal{M}(H) = m_H \mathcal{N}(H)$,

$$\mu_{H_2} = \frac{\mathcal{M}}{m_H \mathcal{N}(H_2)} = \frac{2\mathcal{M}}{m_H \mathcal{N}(H)} = \frac{2\mathcal{M}}{\mathcal{M}(H)} \approx 2.8. \tag{A.1}$$

Note the difference to the mean molecular weight per free particle, μ_p, defined via $\mu_p m_H \mathcal{N} = \mathcal{M}$, where $\mathcal{N} \approx \mathcal{N}(H_2) + \mathcal{N}(He)$ for gas with all H in molecules. As the helium contribution is dominated by ^{4}He, and each H_2 molecule has a weight of two hydrogen atoms, $\mathcal{N}(H_2) = \mathcal{M}(H)/(2m_H)$ and $\mathcal{N}(He) = \mathcal{M}(He)/(4m_H)$. Thus

$$\begin{aligned}
\mu_p &= \frac{\mathcal{M}}{m_H[\mathcal{N}(H_2) + \mathcal{N}(He)]} && \text{(A.2)}\\
&= \frac{\mathcal{M}}{m_H[\mathcal{M}(H)/(2m_H) + \mathcal{M}(He)/(4m_H)]} && \text{(A.3)}\\
&= \frac{\mathcal{M}/\mathcal{M}(H)}{[1/2 + \mathcal{M}(He)/(4\mathcal{M}(H))]} && \text{(A.4)}\\
&\approx 2.37; && \text{(A.5)}
\end{aligned}$$

the classical value of $\mu_p = 2.33$ holds for an abundance ratio $\mathcal{N}(H)/\mathcal{N}(He) = 10$ and a negligible admixture of metals.

Both molecular weights are applied in different contexts. The mean molecular weight per free particle, μ_p, is e.g. used to evaluate the thermal gas pressure, $P = \varrho k_B T/\mu_p$, where T, ϱ, and k_B are the gas temperature, density, and Boltzmann's constant, respectively. The molecular mass per hydrogen molecule, μ_{H_2}, is needed below to derive particle column densities.

A.2. Radiative transfer

A.2.1. Equation of radiative transfer

The intensity emitted by a medium of temperature T and of optical depth τ_ν at the frequency ν is given by the equation of radiative transfer, which reads

$$I_\nu = B_\nu(T)\,(1 - e^{-\tau_\nu}) \tag{A.6}$$

in the case of local thermal equilibrium. Here B_ν is the Planck function. For the optical depth,

$$\tau_\nu = \int \kappa_\nu \varrho \, ds, \tag{A.7}$$

where κ_ν is the (specific) absorption coefficient (i.e., per mass) or dust opacity. If most hydrogen is in molecules, the optical depth can be related to the column density of molecular hydrogen,

$$N_{H_2} = \int n_{H_2} \, ds = \int \frac{\varrho}{\mu_{H_2} m_H} \, ds = \frac{1}{\mu_{H_2} m_H \kappa_\nu} \int \kappa_\nu \varrho \, ds, \tag{A.8}$$

and thus

$$N_{H_2} = \frac{\tau_\nu}{\mu_{H_2} m_H \kappa_\nu}, \tag{A.9}$$

where n_{H_2} is the particle density of hydrogen molecules. This reads in a usable form

$$N_{H_2} = 2.14 \times 10^{25}\ \mathrm{cm^{-2}}\ \tau_\nu \left(\frac{\kappa_\nu}{0.01\ \mathrm{cm^2\,g^{-1}}}\right)^{-1}, \tag{A.10}$$

where the chosen numerical value of κ_ν is characteristic for wavelength $\lambda \approx 1$ mm. As typical H_2 column densities are below 10^{23} cm^2, τ_ν is expected to be by far smaller that 1. Thermal dust emission in the (sub-)millimetre regime is therefore optically thin.

For optically thin conditions the equation of radiative transfer can be simplified:

$$I_\nu \approx B_\nu(T)\,\tau_\nu. \tag{A.11}$$

As the optical depth is related to the column density (Eq. (A.10)), Eq. (A.11) relates the observed intensity to the column density.

A.2.2. The planck function

The Planck function reads

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/(k_B T)} - 1}, \tag{A.12}$$

in which c is the speed of light and h is Planck's constant. In the Rayleigh-Jeans limit, $h\nu \ll k_B T$, this simplifies to

$$B_\nu(T) = \frac{2\nu^2}{c^2} k_B T. \tag{A.13}$$

However, the limiting condition, which reads

$$\lambda \gg 1.44\ \mathrm{mm}\left(\frac{T}{10\ \mathrm{K}}\right)^{-1} \tag{A.14}$$

in useful units, is under typical dust temperatures of about 10 K not fulfilled for observations at about 1 mm wavelength. The exact value of the Planck function is

$$\begin{aligned}
B_\nu(T) &= 1.475 \times 10^{-23}\mathrm{W\ m^{-2}\ Hz^{-1} sr^{-1}} \left(\frac{\nu}{\mathrm{GHz}}\right)^3\\
&\quad \times \frac{1}{e^{0.0048(\nu/\mathrm{GHz})(T/10\ \mathrm{K})^{-1}} - 1} && \text{(A.15)}
\end{aligned}$$

in useful units.

A.3. Observed quantities

The received flux per beam is related to the intensity by

$$F_\nu^{\mathrm{beam}} = \int I_\nu \, P \, d\Omega, \tag{A.16}$$

where P is the normalised power pattern of the telescope (i.e. $\max[P] = 1$). Defining the beam solid angle as the integral

$$\Omega_A = \int P \, d\Omega, \tag{A.17}$$

one can derive a beam-averaged intensity,

$$\langle I_\nu \rangle = F_\nu^{\rm beam}/\Omega_{\rm A}. \tag{A.18}$$

The beam solid angle can be conveniently approximated for telescopes with a beam profile similar to a Gaussian function,

$$P(\theta) = {\rm e}^{-\theta^2/(2\theta_0^2)} \tag{A.19}$$

(where the angle θ gives the distance from the beam center). The parameter θ_0 is related to the half power beam width of the telescope, $\theta_{\rm HPBW}$, via

$$\theta_0 = \frac{\theta_{\rm HPBW}}{\sqrt{8\ln(2)}} \cdot \tag{A.20}$$

For these idealisations the beam solid angle is

$$\begin{aligned}\Omega_{\rm A} &= 2\pi \int_0^\infty P(\theta)\,\theta\,{\rm d}\theta = 2\pi \int_0^\infty {\rm e}^{-\theta^2/(2\theta_0^2)}\theta\,{\rm d}\theta & \text{(A.21)}\\ &= 2\pi\theta_0^2 & \text{(A.22)}\\ &= \frac{\pi}{4\ln(2)}\theta_{\rm HPBW}^2. & \text{(A.23)}\end{aligned}$$

Using the conversion

$$1\ {\rm arcsec} = 4.85\times10^{-6}\ {\rm rad}, \tag{A.24}$$

one obtains

$$\Omega_{\rm A} = 2.665\times10^{-11}\ {\rm sr}\left(\frac{\theta_{\rm HPBW}}{\rm arcsec}\right)^2. \tag{A.25}$$

The parameter $\theta_{\rm HPBW}$ does not need to be the real telescope beam, but depends on the calibration of the data. To give an example, some software packages (e.g., MOPSIC) apply scaling factors to the data (i.e., $F_\nu^{\rm beam}$) when spatially smoothing a map, so that the beam width to that the calibration refers is equal to the spatial resolution of the map after smoothing.

For the idealisations made here the average intensity derived from Eq. (A.18) is a good approximation to the actual intensity only if the source has an extension of the order of the main lobe of the telescope. Otherwise radiation received via the side lobes may have a significant contribution to $F_\nu^{\rm beam}$, and the idealisation of the telescope beam by Gaussian functions is too simple. Inclusion of the side lobes increases $\Omega_{\rm A}$, and thus reduces $\langle I_\nu\rangle$. The nature of the average intensity derived in Eq. (A.18) is therefore comparable to the one of main beam brightness temperatures used in spectroscopic radio observations. Note that, however, spatial filtering of bolometer systems effectively reduces sidelobes. Unfortunately, this reduction depends in details on the mapping pattern and the source source geometry. It can therefore not be included here.

A.4. Derivation of conversion laws

A.4.1. Flux per beam and column density

Equations (A.9, A.11, A.18) relate optical depth, column density, intensity, and the observed flux per beam with each other. Rearrangement yields

$$N_{\rm H_2} = \frac{F_\nu^{\rm beam}}{\Omega_{\rm A}\mu_{\rm H_2}m_{\rm H}\kappa_\nu B_\nu(T)}, \tag{A.26}$$

which reads

$$\begin{aligned}N_{\rm H_2} &= 2.02\times10^{20}\ {\rm cm}^{-2}\left({\rm e}^{1.439(\lambda/{\rm mm})^{-1}(T/10\ {\rm K})^{-1}}-1\right)\left(\frac{\lambda}{\rm mm}\right)^3\\ &\cdot\left(\frac{\kappa_\nu}{0.01\ {\rm cm^2\ g^{-1}}}\right)^{-1}\left(\frac{F_\nu^{\rm beam}}{\rm mJy\ beam^{-1}}\right)\left(\frac{\theta_{\rm HPBW}}{10\ {\rm arcsec}}\right)^{-2} \quad \text{(A.27)}\end{aligned}$$

in useful units.

A.4.2. Flux and mass

The mass is given by the integral of the column densities across the source,

$$M = \mu_{\rm H_2}m_{\rm H}\int N_{\rm H_2}\,{\rm d}A. \tag{A.28}$$

Substitution of Eqs. (A.9, A.11) yields

$$M = \frac{1}{\kappa_\nu B_\nu(T)}\int I_\nu\,{\rm d}A. \tag{A.29}$$

The surface element ${\rm d}A$ is related to the solid angle element ${\rm d}\Omega$ by ${\rm d}A = d^2\,{\rm d}\Omega$, where d is the distance of the source. Thus

$$M = \frac{d^2}{\kappa_\nu B_\nu(T)}\int I_\nu\,{\rm d}\Omega = \frac{d^2 F_\nu}{\kappa_\nu B_\nu(T)}, \tag{A.30}$$

where $F_\nu = \int I_\nu\,{\rm d}\Omega$ is the integrated flux. This reads

$$\begin{aligned}M &= 0.12\ M_\odot\left({\rm e}^{1.439(\lambda/{\rm mm})^{-1}(T/10\ {\rm K})^{-1}}-1\right)\\ &\times\left(\frac{\kappa_\nu}{0.01\ {\rm cm^2\ g^{-1}}}\right)^{-1}\left(\frac{F_\nu}{\rm Jy}\right)\left(\frac{d}{100\ {\rm pc}}\right)^2\left(\frac{\lambda}{\rm mm}\right)^3 \quad \text{(A.31)}\end{aligned}$$

in useful units.

A.5. Dust temperature and opacity

Dust temperatures of order 10 K have been predicted for dense cores since some time. In the past years Zucconi et al. (2001) estimated dust temperatures in the range 11 to 6 K for solar neighborhood cores. The gas and dust temperature in these should be similar (Goldsmith 2001; also see Galli et al. 2002). Gas temperatures of order 10 K are indeed common for dense cores (e.g., Tafalla et al. 2004). Such temperatures even prevail in dense cores residing in molecular clouds forming stellar clusters (i.e., in Perseus; Rosolowsky et al. 2007). Finally, Schnee et al. (2007b) recently mapped the dust temperature distribution in TMC-1C using data partially also included in the present work. They derived dust temperatures of 13 to 5 K, similar to gas temperatures recently derived for L1544 (Crapsi et al. 2007). These temperatures are lower than those typically inferred from IRAS ($\gtrsim$15 K; Schnee et al. 2005) and ISO data ($\gtrsim$10 K; Lehtinen et al. 2005), possibly because these missions primarily probed very extended emission because of large beams. We therefore here adopt a temperature of 10 K in absence of protostellar heating.

Figure A.1 illustrates the uncertainty in mass estimates due to uncertainties in the temperature (based on Eq. (A.31)). For MAMBO this shows that for actual dust temperatures of 12 to 8 K our assumption of a dust temperature of 10 K will lead to an error in the mass estimate of factors 0.7 to 1.5, respectively. These increase to factors of 0.6 to 3 for temperatures of 14 to 6 K. The uncertainty significantly decreases with increasing wavelength.

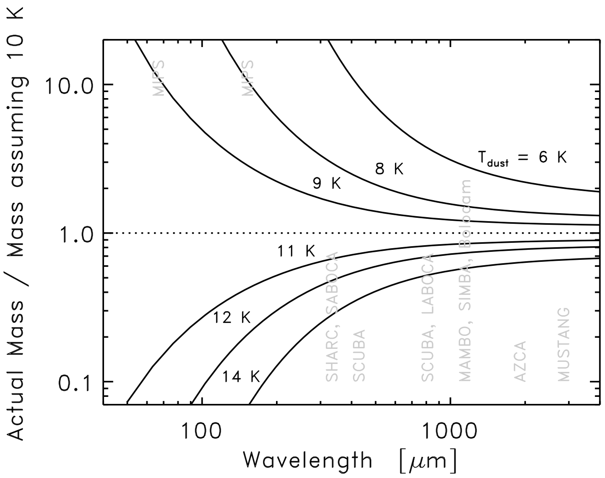

Fig. A.1. The ratio of the actual mass to the one derived when assuming a dust temperature of 10 K. *Solid lines* indicate the ratio for various actual dust temperatures. *Vertical labels* indicate the observing wavelength of several existing and upcoming bolometer arrays (for completeness also including Spitzer MIPS). The related uncertainty in mass estimates decreases significantly with increasing wavelength.

Dust opacities are presently not well constrained. We here adopt dust opacities from Ossenkopf & Henning (1994) that hold for dust with thin ice mantles coagulating for 10^5 yr at an H-density of 10^6 cm^{-3}, respectively (keeping the product of time and density constant) for 10^6 yr at an H_2-density of 5×10^4 cm^{-3}, which appear to be reasonable conditions. At the MAMBO wavelength of 1.2 mm their opacity of 0.01 cm^2 per gram of interstellar matter is in between values used and suggested by other groups (from 0.005 cm^2 g^{-1} Motte et al. 1998) to 0.02 cm^2 g^{-1}, Krügel & Siebenmorgen 1994).

The opacities quoted above are consistent with observational opacity constraints. van der Tak et al. (1999) conclude that only coagulated dust from Ossenkopf & Henning (1994) provides a match to their extensive dust and molecular line emission data set. They probed a massive young star with a heated envelope though. Anyway, Shirley et al. (submitted) derive the same for a less luminous YSO in B335.

A.6. Standard c2d conversion factors

A.6.1. Conversion from dust emission

Table A.1 summarises the c2d standard conversion factors for masses and column densities from dust emission. The wavelength-dependent dust opacities are from Ossenkopf & Henning (1994) and hold for dust with thin ice mantles coagulating for 10^5 yr at an H-density of 10^6 cm^{-3}. We adopt a dust temperature of 10 K. This choice for the dust temperature and opacity are the standard assumptions made by the c2d collaboration. The values listed in Table A.1 are thus thought to serve as a standard reference within the collaboration.

A.6.2. Conversion to extinction

The c2d standard conversion factor between column densities and visual extinction,

$$N_{H_2} = 9.4 \times 10^{20}\ \mathrm{cm}^{-2}\,(A_V/\mathrm{mag})\,, \qquad (A.32)$$

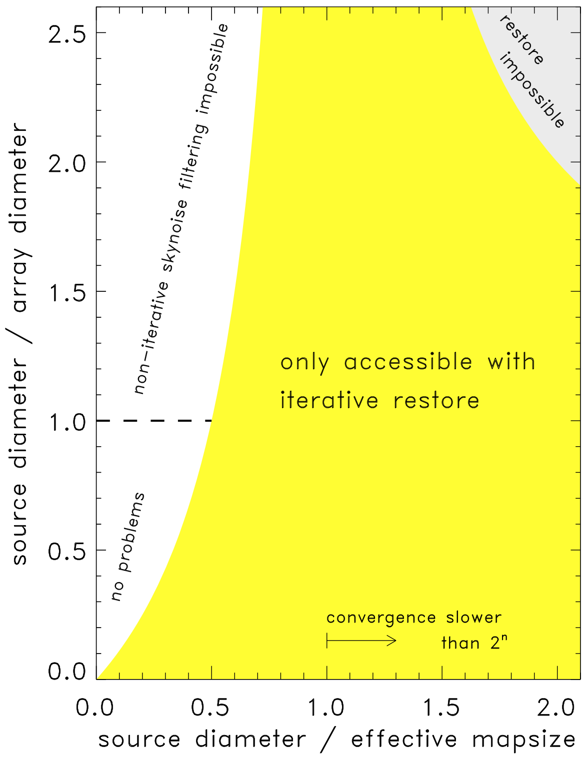

Fig. B.1. Summary of mapping parameter spaces accessible by various techniques. The *non-highlighted area* at $d_{source}/\ell_{eff} < 1$ is accessible using conventional techniques. *This is the realm in which any past and present bolometer arrays, including total power devices like SCUBA II and LABOCA, operate.* If one whishes to remove skynoise, and does not use an iterative approach for this, this further limits the accessible parameter space to $d_{source}/d_{array} < 1$. Failure to comply with these limits results in artifacts of various kind. The *area with light highlighting* is accessible when using our iterative restore algorithm, which by design also includes iterative skynoise filtering. This significantly increases the accessible mapping parameter space. However, as indicated by the *arrow*, the convergence speed decreases when increasing d_{source}/ℓ_{eff}. Maps occupying the *area with strong highlighting* cannot be reduced without artifacts.

is taken from Bohlin et al. (1978). They combined measurements of H_2 and HI from the Copernicus satellite for lightly reddened stars to get

$$\langle [N_{\mathrm{HI}} + 2N_{\mathrm{H_2}}]/E(B-V) \rangle = 5.8 \times 10^{21}\ \mathrm{cm}^{-2}\ \mathrm{mag}^{-1}\,. \qquad (A.33)$$

For a standard total-to-selective extinction ratio $R_V = A_V/E(B-V) = 3.1$ this yields the above conversion factor.

Appendix B: Data reduction details

B.1. Mapping parameter space

The equations on limitations of data reduction approaches, Eqs. (1–3), effectively span parameter spaces accessible by different reduction methods. We explore these limits in Fig. B.1.

For brevity, we express the effective map size, $\ell_{scan} + \ell_{chop}$, by ℓ_{eff} in the following. The vertical limit at $d_{source}/d_{array} = 1$ directly follows from Eq. (3). In principle, it continues to $d_{source}/\ell_{eff} \to \infty$. However, Eq. (2) anyway limits the realm of unproblematic data reduction along this axis. To obtain an expression for this limit, we divide Eq. (2) by d_{array} and solve for